

Lab program 3.c - Codes:Blocks 25.03

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```
1  #include<stdio.h>
2  #define MAX5
3  int stack[MAX5];
4  int top=-1;
5  void push(int value){
6      if(top==MAX5-1)
7          printf("stack Overflow\n");
8      else{
9          top++;
10         stack[top] = value;
11         printf("pushed %d\n",value);
12     }
13 }
14 }
15 void pop(){
16     if(top==--1)
17         printf("stack underflow\n");
18     else{
19         printf("popped %d/n",stack[top]);
20         top--;
21     }
22 }
23 void display(){
24     if(top==--1)
25         printf("stack is empty\n");
26     else{
27         printf("stack elements:\n");
28         for(int i=top;i>=0;i--);
29         printf("%d\n",stack[i]);
30     }
31 }
32 }
33 int main(){
```

C:\Users\Sevanthi-v\Downloads\Lab program 3.cC/C++Windows (CR+LF)WINDOWS-1252Line 29, Col 32, Pos 512InsertRead/Writedefault

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```
7     printf("stack Overflow\n");
8 } else{
9     top++;
10    stack[top] = value;
11    printf("pushed %d\n",value);
12
13 }
14 }
15 void pop(){
16 if(top==--1)
17     printf("stack underflow\n");
18 } else{
19     printf("popped %d\n",stack[top]);
20     top--;
21 }
22 }
23 void display(){
24 if(top==--1)
25     printf("stack is empty\n");
26 } else{
27     printf("stack elements:\n");
28     for(int i=top;i>=0;i--);
29     printf("%d\n",stack[i]);
30 }
31 }
32 }
33 int main(){
34     push(10);
35     push(20);
36     pop();
37 }
38 }
```

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```
lab program DS1.c - Code::Blocks 25.03
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Start here X Lab program 3.c X lab program DS1.c X lab program DS2.c X DS lab 9.c X

1
2  /*implement basic array operations:insertion,deletion,searching,and traversal*/
3  #include<stdio.h>
4  int main(){
5      int arr[100],n,i,pos,value,choice,found;
6      printf("Enter number of element:");
7      scanf("%d",&n);
8      printf("Enter element;\n");
9      for(i=0;i<n;i++){
10         scanf("%d",&arr[i]);
11     }
12     do{
13         printf("\n---MENU---\n");
14         printf("1.Traversal\n2.Insertion\n3.Deletion\n4.searching\n5.Exit\n");
15         printf("Enter your choice");
16         scanf("%d",&choice);
17         case1:
18             printf("Array elements are:");
19             for(i=0;i<n;i++){
20                 printf("%d",arr[i]);
21             }
22             break;
23         case2:
24             printf("Enter positive and value:");
25             scanf("%d %d",&pos,&value);
26             for(i=0;i>=pos;i--){
27                 arr[i]=arr[i-1];
28             }
29             arr[pos-1]=value;
30             n++;
31             break;
32         case3:
33             printf("Enter position to delete:");
34             scanf("%d",&pos);
35             for(i=pos-1;i<n-1;i++){
```

lab program DS1.c - Code::Blocks 25.03

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```
30     n++;
31     break;
32 case3:
33     printf("Enter position to delete:");
34     scanf("%d",pos);
35     for(i=pos-1;i<n-1;i++){
36         arr[i]==arr[i+1];
37     }
38     n--;
39     break;
40 case4:
41     printf("Enter element to search");
42     scanf("%d",value);
43     found=0;
44     for(i=0;i<n;i++){
45         if(arr[i]== value){
46             printf("element found position:",i+1);
47             found=1;
48             break;
49         }
50     }
51     if(found==0)
52         printf("element not found/n");
53     break;
54 case5:
55     printf("Exiting.../n");
56     break;
57 default:
58     printf("invalid choice!/n");
59 }
60 while(choice !=5);
61 return 0;
62 }
63
```

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```
lab program DS2.c - Code::Blocks 25.03
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Start here x Lab program 3.c x lab program DS1.c x lab program DS2.c x DS lab 9.c x

1
2 #include<stdio.h>
3
4 int main(){
5     int matrix[10][10],transpose[50][3],triplet[50][3];
6     int i,j,m,n,k=1;
7
8     printf(" enter rows and coloums:");
9     scanf("%d %d",&m,&n);
10    printf(" enter matrix elements");
11    for(i=0;i<m;i++){
12        for(j=0;j<n;j++){
13            scanf("%d",&matrix[i][j]);
14        }
15    }
16
17    triplet[0][0]=m;
18    triplet[0][1]=n;
19    triplet[0][2]=0;
20    for(i=0;i<m;i++){
21        for(j=0;j<n;j++){
22            if(matrix[i][j]!=0){
23                triplet[k][0]=i;
24                triplet[k][1]=j;
25                triplet[k][2]=matrix[i][j];
26                k++;
27                triplet[0][2]++;
28            }
29        }
30    }
31    printf("\nTriple Representation:");
32    for(i=0;i<k;i++){
33        printf("%d %d %d\n",triplet[i][0],triplet[i][1],triplet[i][2]);
34    }
35 }
```

C:\Users\Sevanthi-v\Downloads\lab program DS2.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 23, Col 29, Pos 453 Insert Read/Write default

```
lab program DS2.c - Code::Blocks 25.03
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Start here x Lab program 3.c x lab program DS1.c x lab program DS2.c x DS lab 9.c x
24 triplet[k][1]=j;
25 triplet[k][2]=matrix[i][j];
26 k++;
27 triplet[0][2]++;
28 }
29 }
30 }
31 printf("\nTriple Representation:");
32 for(i=0;i<k;i++){
33     printf("%d %d %d\n", triplet[i][0], triplet[i][1], triplet[i][2]);
34 }
35 }
36
37 transpose[0][0]=triplet[0][1];
38 transpose[0][1]=triplet[0][0];
39 transpose[0][2]=triplet[0][2];
40 k=1;
41 for(i=0;i<n;i++){
42     for(j=1;j<triplet[0][2];j++){
43         if(triplet[j][1]==i){
44             transpose[k][0]=triplet[j][1];
45             transpose[k][1]=triplet[j][0];
46             transpose[k][2]=triplet[j][2];
47             k++;
48         }
49     }
50 }
51 printf("\nTranspose triplet/n");
52 for(i=0;i<k;i++){
53     printf("%d %d %d", transpose[i][0], transpose[i][1], transpose[i][2]);
54 }
55 return 0;
56 }
57
```

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C/C++

Windows (CR+LF)

WINDOWS-1252

Line 23, Col 29, Pos 453

Insert

Read/Write

default



DS lab 9.c - Code::Blocks 25.03

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Start here x Lab program 3.c x lab program DS1.c x *lab program DS2.c x DS lab 9.c x

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node* next;
7  };
8
9
10 struct Node* createNode(int data) {
11     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
12
13     if (newNode == NULL) {
14         printf("Memory allocation failed\n");
15         exit(1);
16     }
17
18     newNode->data = data;
19     newNode->next = NULL;
20     return newNode;
21 }
22
23
24 void insertAtBeginning(struct Node** head, int data) {
25     struct Node* newNode = createNode(data);
26     newNode->next = *head;
27     *head = newNode;
28 }
29
30
31 void insertAtEnd(struct Node** head, int data) {
32     struct Node* newNode = createNode(data);
33
34     if (*head == NULL) {
35         *head = newNode;
```

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Start here x Lab program 3.c x lab program DS1.c x lab program DS2.c x DS lab 9.c x

```
34 if (*head == NULL) {
35     *head = newNode;
36     return;
37 }
38
39 struct Node* temp = *head;
40 while (temp->next != NULL) {
41     temp = temp->next;
42 }
43
44 temp->next = newNode;
45 }
46
47
48 void insertAtPosition(struct Node** head, int data, int position) {
49     if (position < 0) {
50         printf("Invalid position\n");
51         return;
52     }
53
54     if (position == 0) {
55         insertAtBeginning(head, data);
56         return;
57     }
58
59     struct Node* newNode = createNode(data);
60     struct Node* temp = *head;
61
62     for (int i = 0; temp != NULL && i < position - 1; i++) {
63         temp = temp->next;
64     }
65
66     if (temp == NULL) {
67         printf("Position out of range\n");
68         free(newNode);
```

C:\Users\Sevanthi-v\Downloads\DS lab 9.cC/C++Windows (CR+LF)WINDOWS-1252Line 62, Col 1, Pos 1202InsertRead/Writedefault

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Start here X Lab program 3.c X Lab program DS1.c X Lab program DS2.c X DS lab 9.c X

```
67     printf("Position out of range\n");
68     free(newNode);
69     return;
70 }
71
72     newNode->next = temp->next;
73     temp->next = newNode;
74 }
75
76
77 void deleteFromBeginning(struct Node** head) {
78     if (*head == NULL) return;
79
80
81     struct Node* temp = *head;
82     *head = (*head)->next;
83     free(temp);
84 }
85
86 void deleteFromEnd(struct Node** head) {
87     if (*head == NULL) return;
88
89     if ((*head)->next == NULL) {
90         free(*head);
91         *head = NULL;
92         return;
93     }
94
95     struct Node* temp = *head;
96
97     while (temp->next->next != NULL) {
98         temp = temp->next;
99     }
100
101     free(temp->next);
```

C:\Users\Sevanthi-v\Downloads\DS lab 9.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 62, Col 1, Pos 1202 Insert Read/Write default

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Start here x Lab program 3.c x Lab program DS1.c x Lab program DS2.c x DS lab 9.c x

```
100
101     free(temp->next);
102     temp->next = NULL;
103 }
104
105
106 void deleteFromPosition(struct Node** head, int position) {
107     if (*head == NULL || position < 0) return;
108
109     if (position == 0) {
110         deleteFromBeginning(head);
111         return;
112     }
113
114     struct Node* temp = *head;
115
116     for (int i = 0; temp != NULL && i < position - 1; i++) {
117         temp = temp->next;
118     }
119
120     if (temp == NULL || temp->next == NULL) {
121         printf("Position out of range\n");
122         return;
123     }
124
125     struct Node* nodeToDelete = temp->next;
126     temp->next = nodeToDelete->next;
127     free(nodeToDelete);
128 }
129
130
131 void traverse(struct Node* head) {
132     struct Node* temp = head;
133
134     while (temp != NULL) {
```

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Start here x Lab program 3.c x Lab program DS1.c x Lab program DS2.c x DS lab 9.c x

```
133
134     while (temp != NULL) {
135         printf("%d -> ", temp->data);
136         temp = temp->next;
137     }
138
139     printf("NULL\n");
140 }
141
142
143 int search(struct Node* head, int target) {
144     struct Node* temp = head;
145
146     while (temp != NULL) {
147         if (temp->data == target)
148             return 1;
149         temp = temp->next;
150     }
151
152     return 0;
153 }
154
155
156 void freeList(struct Node* head) {
157     struct Node* temp;
158
159     while (head != NULL) {
160         temp = head;
161         head = head->next;
162         free(temp);
163     }
164 }
165
166 int main() {
167     struct Node* head = NULL;
```

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```
163     }
164 }
165
166 int main() {
167     struct Node* head = NULL;
168
169     insertAtEnd(&head, 10);
170     insertAtEnd(&head, 20);
171     insertAtEnd(&head, 30);
172     traverse(head);
173
174     insertAtBeginning(&head, 5);
175     traverse(head);
176
177     insertAtPosition(&head, 25, 2);
178     traverse(head);
179
180     deleteFromBeginning(&head);
181     traverse(head);
182
183     deleteFromEnd(&head);
184     traverse(head);
185
186     deleteFromPosition(&head, 1);
187     traverse(head);
188
189     printf("List contains 20: %s\n", search(head, 20) ? "Yes" : "No");
190     printf("List contains 40: %s\n", search(head, 40) ? "Yes" : "No");
191
192     freeList(head);
193
194     return 0;
195 }
196
```

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